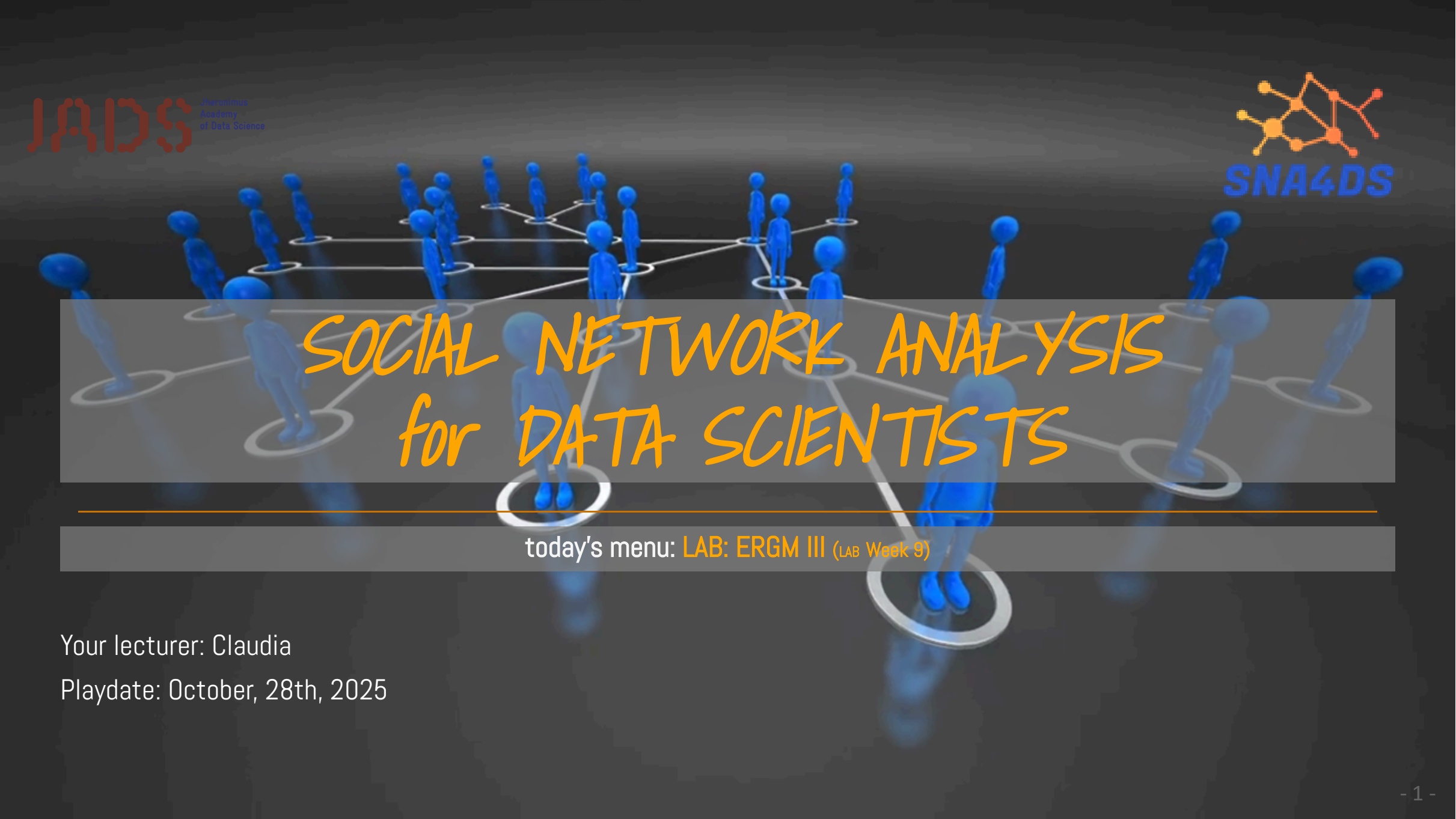




SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: ERGM III (LAB Week 9)

Your lecturer: Claudia

Playdate: October, 28th, 2025

Turn your
cameras on!



Menu

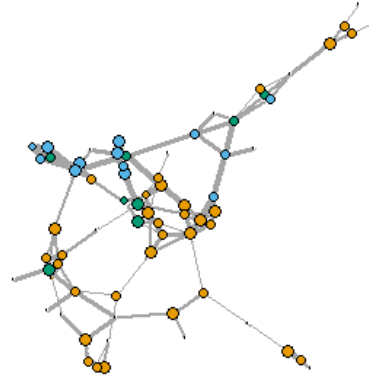


- Super Lab - Real Time ERGM!
- Final Recommendations
- ERGMs Qs & As

Real Time ERGM

Let's Research JADS Data! :)

Friendship by Nationality DS and closeness



```
setwd("C:/Users/...")
```

```
load("JADSmaster2022.Rdata")
```

60:00



SNA4DS

Final Recomendations

Remember



- Network outcome variable
- Always use structural terms
- Handling large networks
- Handling complex cases
- Seven step flow approach

Network outcome variable



You already know that ERGMs work with a network outcome variable.

Make sure you are developing your theory accordingly!

Some examples of Hypotheses

- It is more likely that people of the same age are friends
- More popular scholars prefer to collaborate with other popular scholars
- Companies want to trade with other companies in the same sector

Thinking in Variables:



It is more likely that people of the same age are friends

- outcome: friendship
- predictor: age (numeric)

More popular scholars prefer to collaborate with other popular scholars

- outcome: collaboration
- predictor: popularity (numeric/ordinal)

Companies want to trade with other companies in the same sector

- outcome: trade
- predictor: sector (categorical)

Always use structural terms -



Even if your hypotheses do not point you to structural terms, they still must be in the model (2 H at least).

Two reasons

- Otherwise it is just like a logistic regression
- The model won't work well
- It's compulsory

ERGMs explain relationships.

Allow them to do their job

Handling large networks



Ideally, for this assignment, you should stay below 500 nodes

Handling large networks



Ideally, for this assignment, you should stay below 500 nodes

Why?

Handling large networks



Ideally, for this assignment, you should stay below 500 nodes

Why?

Just because it takes too much time to run these models and you have a tight deadline.

Handling large networks



If your project requires large networks (even around 500)

1. use control ergm to optimize the simulation parameters
2. constraint the model so that the simulation gets a little faster
3. run it on a server
4. parallelize

Handling complex cases



Some of you are re-constructing fairly complex networks

Make sure that you know what the terms you insert in the model mean

It is better to have a simple project than super fancy attempts that nobody can understand.

Also, make sure you don't do too many transformations with your data before putting them into the model.

SUPER DISCLAIMER

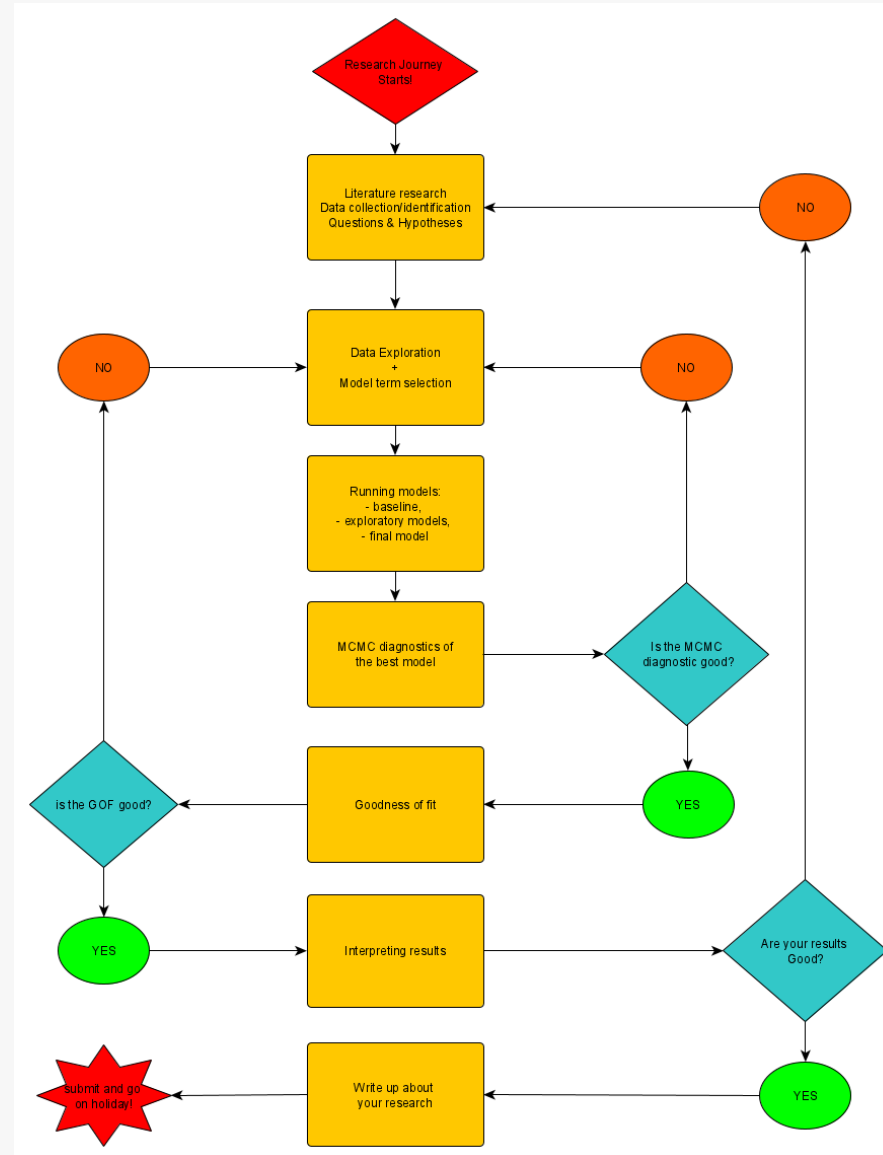


WE DON'T CARE ABOUT STARS

- Significant effects + good MCMC diagnostics + good GOF = AWESOME

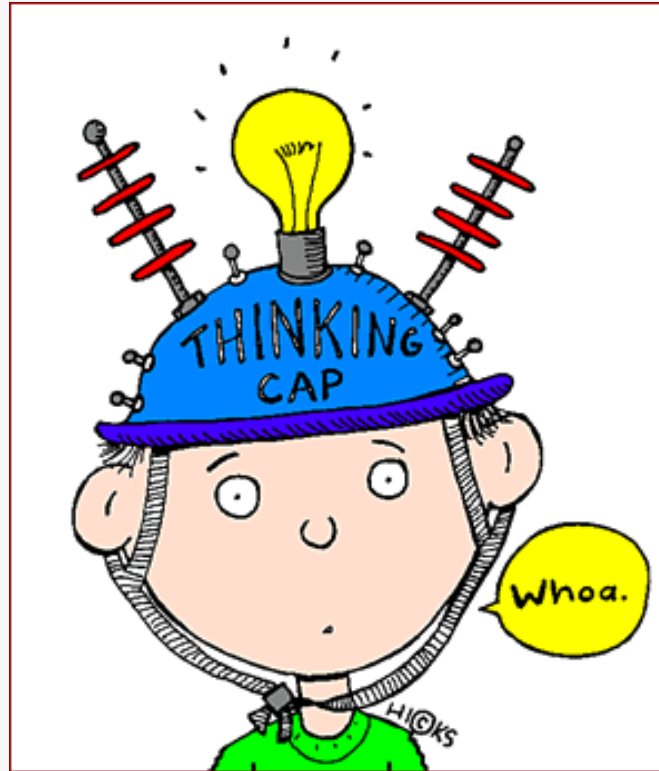
BUT

- Significant effects + bad MCMC diagnostics + bad GOF = bad mark
- NON-Significant effects + good MCMC diagnostics + good GOF = good mark (IF YOU CAN EXPLAIN WHY)



ERGMS Qs & As

Final point: Think carefully about your research questions!



Good Luck backing your Projects!



